



Interlace: A Shared Architecture for Language and Iterative Computation

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Abstract. Interlace is a shared language architecture designed to improve learned capability per stored parameter and training budget. It combines bounded rational features, content-selected attention, gated delta memory and repeated shared depth. A 29,814,640-parameter model trained from random initialization is compared with a 29,921,280-parameter dense Transformer under the same BF16 joint training recipe. The combined model has 9.86% lower official test loss at the same training budget. It reaches the fixed 5.5-nat validation-monitor target in 276.52 seconds versus 806.91 seconds for the dense control. Longer-program stream accuracy is 80.47% versus 4.69% for the dense control. These measurements motivate testing whether the architecture can reduce the cost of reaching useful model quality. Both models use the same synchronous stream interface; the comparison measures the combined backbone rather than any isolated component. Results cover one seed, WikiText-2 and bounded prefix-addition programs. The released FP8 checkpoint provides a compact reference implementation; Flow-style refinement is an experimental extension. We specify a staged validation program to test repeatability, component contributions and transfer to larger data and task budgets.

1 Introduction

A model can increase its effective computation without storing a separate parameter set for every computational stage. It can also combine selective access to past representations with a recurrent memory that accumulates information over time. The research question is whether these mechanisms, trained together, produce more useful capability at a given model size and training cost.

Interlace integrates four backbone mechanisms: a bounded rational feed-forward transformation, content-selected attention, gated delta memory, and a repeated shared stack. Two additional interfaces expose computation in different ways: synchronous streams produce intermediate states alongside external inputs, while Flow refinement revises a complete candidate solution. These interfaces share the language backbone; no external fact store or teacher supplies their answers.

The central measurement is time to a fixed language-quality target. We report that quantity alongside final held-out loss, stored parameters and inference latency. Repeated seeds and controlled architectural changes are needed to understand how these quantities depend on the combined mechanisms.

The contribution is the implemented combination, its explicit mathematical contracts and a controlled evaluation at approximately 30 million parameters. The paper separates architectural quality, training cost, inference cost and representation precision. A parameter-matched dense control tests the combined backbone under a common recipe. It does not isolate the contribution of each component, which requires further ablations.

2 Architecture

The model uses width 640, feed-forward width 1,536, eight heads, a 4,096-token byte-level BPE vocabulary and a 128-row context. Six residual blocks alternate three attention and three memory blocks. The same stack is applied twice, giving twelve block evaluations with six stored block parameter sets. Learned role, mode, position and loop embeddings are included in the total parameter count. Text uses a tied input/output vocabulary matrix. Structured refinement adds a ten-category output head.

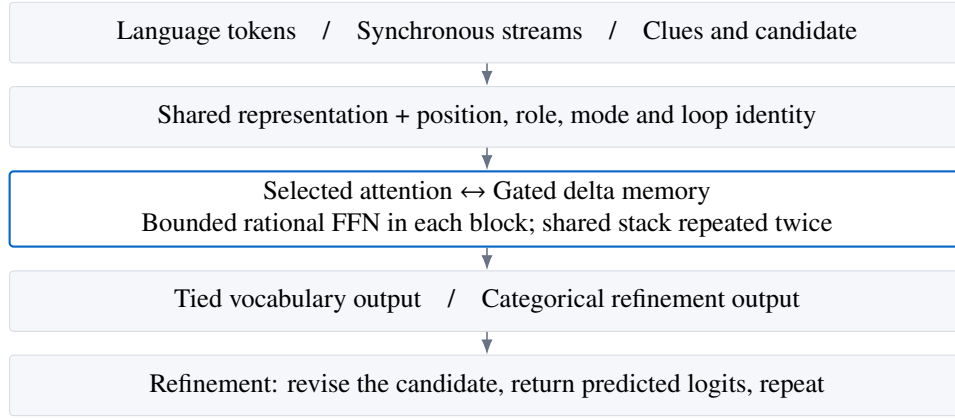


Figure 1: A common parameter set supports three interfaces. Shared depth repeats latent computation; refinement explicitly revises predicted candidate states.

2.1 Bounded rational features

Rational architectures motivate expressive feature interactions beyond a fixed scalar activation [1, 2]. The feed-forward block uses paired learned projections and a positive quadratic denominator:

$$(a, b) = W_{\text{up}}x, \quad r(a, b) = \frac{a(1+b)}{1+a^2+b^2}, \quad \text{FFN}(x) = W_{\text{down}}r(a, b). \quad (1)$$

For fixed b , maximizing over a gives $|r| \leq |1+b|/(2\sqrt{1+b^2}) \leq 1/\sqrt{2}$. The gate therefore bounds the interaction without a denominator floor. Its derivatives are

$$\frac{\partial r}{\partial a} = \frac{(1+b)(1+b^2-a^2)}{(1+a^2+b^2)^2}, \quad \frac{\partial r}{\partial b} = \frac{a(1+a^2-b^2-2b)}{(1+a^2+b^2)^2}. \quad (2)$$

The implementation evaluates an algebraically equivalent scaled expression using $s = \max(1, |a|, |b|)$ to avoid overflow. The explicit parameterization is part of this architecture; rational networks themselves have prior art. A bounded gate alone does not imply greater knowledge capacity.

2.2 Selected attention and delta memory

Attention selects two completed eight-row source blocks using content summaries, together with the current block. Exact softmax attention operates within the selected mask. This follows the selective-access direction of Monodratic [9]; the present implementation does not reproduce its complete index. Text and stream queries cannot read future rows. Flow can read the full current candidate because all candidate positions are available at inference.

Memory complements selective reads with a gated delta recurrence [3]. Each head has 16 key coordinates and 80 value coordinates. With normalized keys and queries and learned decay α_t and write strength β_t ,

$$\tilde{S}_t = \text{diag}(\alpha_t)S_{t-1}, \quad e_t = v_t - k_t^T \tilde{S}_t, \quad S_t = \tilde{S}_t + \beta_t k_t e_t, \quad o_t = q_t^T S_t. \quad (3)$$

The memory input is the mean of the current row's role representations. Its output is gated separately for each role. This state is transient computation; it is not a persistent retrieval database.

2.3 Synchronous streams

The stream interface receives external input digits and predicts thought and output tokens over successive ticks. The initial task is prefix addition modulo ten, which provides exact intermediate-state and final-answer labels. During evaluation, thought and output tokens are generated freely over the full vocabulary. No reference intermediate state is supplied to inference.

2.4 Experimental candidate refinement

The refinement interface adapts Flow Reasoning Models [4]. Given clues c , categorical endpoint y , Gaussian noise z and time t , training uses $x_t = (1 - t)z + ty$. A denoiser predicts category logits $\ell_\theta(c, x_t, t, f)$, with either zero feedback or detached logits from a previous prediction. A masked categorical cross-entropy supervises valid program positions. One quarter of training times are zero, where the candidate contains no target contribution.

At inference the candidate begins as noise. At each of K uniform steps,

$$p_i = \text{softmax}(\ell_\theta(c, x_i, t_i, f_i)), \quad x_{i+1} = x_i + \frac{p_i - x_i}{K - i}, \quad f_{i+1} = \ell_\theta(c, x_i, t_i, f_i). \quad (4)$$

All candidate positions can change. The evaluated checkpoint uses ordinary self-conditioning; trajectory-based Fixed-Point Forcing is not part of its selected training recipe. Re-noise consistency measures stability of a prediction and must not be interpreted as a correctness certificate.

3 Experimental method

3.1 Data, training and controls

All models begin from random initialization. The language corpus is WikiText-2 raw [8], with a tokenizer fitted only on its 629 prepared training documents. Each joint run contains 16,384 updates: 8,192 language updates with 4,194,304 sampled target tokens, 4,096 stream updates and 4,096 Flow updates. Training programs have lengths two through six. Sequence hashes separate training, validation and test programs. The fixed final tests contain 256 short and 128 longer programs, with lengths eight through ten in the latter group.

Hidden matrices use NorMuon updates with five Polar Express iterations, momentum 0.7, row second moments and update RMS 0.2 [5, 6]. Matrix learning rate is 0.006; embeddings and auxiliary parameters use AdamW at 0.001. Weight decay is 0.01. The schedule warms up for 64 updates, remains constant through 80% of training and decays linearly to one tenth. The architecture comparison holds the training precision at BF16 and uses the same seed, batch order, task mixture, optimizer and schedule.

The dense control has six unique full-attention/SwiGLU blocks [7], applied once, with width 640 and hidden width 1,504. It retains the same tied embedding and text/stream/Flow adapters so that task access is comparable. Its 29,921,280 parameters exceed the combined model by only 0.36%. It has no delta-memory blocks, rational gates, content-selected mask or repeated shared stack. This is a matched comparison of two concrete architectures under a common recipe, not an independently tuned search over dense models.

3.2 Measurement and reproducibility

Language quality uses all 2,488 official validation and 2,844 official test windows. The test corpus is assembled from the same pinned source revision; its documents have no title, whole-document or normalized paragraph overlap of at least 80 characters with the prepared training and validation data. The dense recipe is fixed before opening its test results. The baseline is a new comparison against an already evaluated model, not a blind model-selection competition.

We report exact sequence and final-answer accuracy separately. Longer inputs test length generalization beyond training. Disabling generated thought and removing feedback are inference interventions, not retrained ablations. Numerical checks compare rational and dense feed-forward values and gradients, attention masks and memory recurrence against independent host calculations.

Elapsed training cost includes initialization, compilation, data handling, validation, checkpoint handling, numerical retries and interrupted execution. The combined BF16 run includes recovery after an overflow; the original failed run remains in the record. First observed monitor crossings at 6.0, 5.5 and 5.0 nats/token are measured at fixed 512-update intervals, without interpolating an unobserved crossing. Inference measurements synchronize completed device work and use eighteen repetitions across two fresh processes per model, in a counterbalanced order, after separately reported first executions. Generation uses the same uncached implementation in both arms. Measurements use one local accelerator and a bounded process working set;

hardware, software versions, raw times, resource records and checksums accompany the technical supplement. Distributed performance is outside this evaluation.

4 Results

4.1 Architecture comparison

Measurement	Interlace BF16	Dense BF16
Parameters, millions	29.815	29.921
Full validation loss, nats/token	4.7187	5.2278
Official test loss, nats/token	4.7339	5.2516
Official test bits/byte	1.9511	2.1645
Held-out stream answers (%)	100.00	100.00
Longer-program stream answers (%)	80.47	4.69
Held-out eight-pass Flow, exact (%)	0.00	0.39
Total training execution, minutes	22.03	22.26

Table 1: Equal update and language-token budgets, common BF16 precision and one seed. All task scores come from the same final checkpoint in each arm.

At this fixed budget, the combined model has 9.86% lower official language-test loss. Total training execution is 1.04% shorter; this cost includes the disclosed interrupted work. On longer programs, its stream answer accuracy is 80.47% versus 4.69%. These are observed differences between the two trained models, not evidence that every architectural component improves performance. Additional seeds, component ablations and independently tuned dense controls remain necessary.

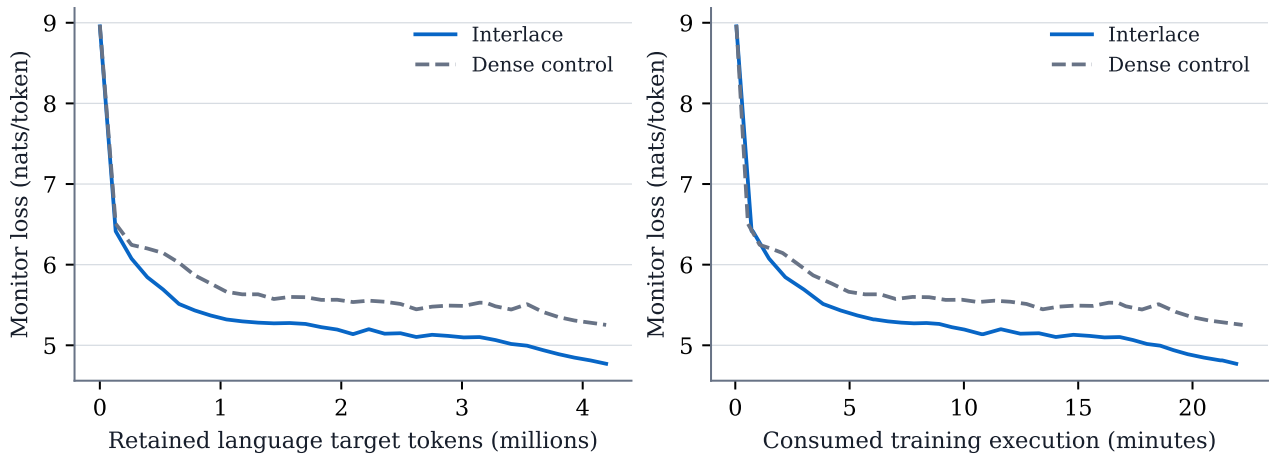


Figure 2: Language learning at matched retained token budgets and measured execution time. Curves use the same fixed 32-window validation monitor; full-split results are reported in the table. Retried work is included in time.

First observed monitor target, seconds	Interlace BF16	Dense BF16
6.0 nats/token	131.29	204.37
5.5 nats/token	276.52	806.91
5.0 nats/token	1115.28	Not reached

Table 2: Time to a fixed monitor quality, sampled every 512 updates. Each time includes consumed training execution before the observed crossing.

Inference latency, median	Interlace BF16	Dense BF16
128-token forward, batch 1 (ms)	6.14	2.11
Generate 32 tokens, batch 1 (s)	0.11	0.04
Eight-pass Flow, batch 8 (ms)	30.60	10.63

Table 3: Eighteen completed-device repetitions per workload across two counterbalanced processes per model, after separately recorded first executions. Both models use BF16 and the same uncached generation procedure. These are local implementation measurements.

The dense control has lower latency in all three reported inference workloads. The measured improvement in held-out quality and earlier monitor crossings therefore accompanies higher inference latency in this implementation.

4.2 Reference checkpoint and precision

The reference checkpoint uses FP8 weights and projection operands, with wider working tensors. Its complete learned tensor payload, including padding and scales, is 48.44% smaller than an equal-parameter BF16 representation; the weight file is 30.76 MB. Relative language loss differs by +0.49% on validation and +0.56% on the official test against the same architecture trained in BF16. Both are below the registered 1% precision threshold for this pair. This is a secondary representation result, separate from the architectural comparison above.

Free-running streams produce 100.00% correct held-out answers and 92.97% on longer programs. Clamping the thought stream to idle lowers held-out accuracy to 9.77%. These results show a learned intermediate-state dependency on the evaluated task. Since the dense control uses the same stream interface, its comparison with Interlace cannot isolate the stream mechanism as the cause of the accuracy difference.

Eight-pass Flow produces 1.95% exact held-out state sequences, versus 0.39% after one pass. It produces no exact longer-program sequences in this test. The implemented mechanism supports candidate revision, but reliable refinement remains an open capability gate. Language samples exhibit repetition and fragmented continuations; no general question-answering benchmark has been passed.

5 System card

Field	Current specification
Developer	Deyan Todorov, Misul
Model	Approximately 30M parameters; shared language, stream and refinement backbone
Initialization	Random; no pretrained checkpoint, teacher distillation or external answer store
Language data	WikiText-2 raw; train-only byte-level BPE, vocabulary 4,096
Structured data	Synthetic prefix-sum programs modulo ten; training lengths 2–6
Context	Up to 128 rows; one language stream or three synchronous roles
Outputs	Autoregressive tokens, generated intermediate/final program tokens, or categorical candidate states
Intended use	Architecture research, controlled capability evaluation and reproducible inference experiments
Current limits	Fragmented text; weak Flow exact accuracy; no demonstrated general instruction following or factual QA
Training state	BF16 working parameters and optimizer; separately counted from the compact released weights
Release status	Research checkpoint and reproducibility package; Apache-2.0 license
Evaluation scope	One training seed for the 30M dense comparison; fixed short context and bounded synthetic programs

Table 4: The system card describes the evaluated checkpoint, rather than an anticipated larger model.

The current model has not undergone a comprehensive bias, memorization, robustness or misuse evaluation. Generated text can be incorrect or reproduce patterns in the training corpus. The architecture does not implement tool use, autonomous actions or a persistent external memory. The reproducibility package preserves failed examples alongside successful outputs and includes model, tokenizer and source digests.

6 Validation program

The next study tests whether the observed learning advantage survives new seeds, architectural controls and broader workloads. The following are prospective decision criteria, not additional results. Protocols, data splits, resource caps and analysis scripts will be frozen before confirmation runs.

Repeatability. Run three fresh paired seeds with unchanged recipes. Use paired summaries of full validation loss and first observed time to the fixed 5.5-nat monitor target; retain every seed, interruption and failed crossing. Advance if all three pairs improve full validation loss and median time to target is at least 10% lower. If the target is not reached within the common cap, record a non-crossing rather than extrapolating time. Open the final test only after the recipes are fixed.

Mechanism and baseline. Screen rational-gate, attention-selection, delta-memory and shared-depth ablations under the same data budget, reporting parameter count and actual execution cost. Add a dense shared-depth control to distinguish depth reuse from the other changes. Give both architecture families equal validation-only optimizer-tuning budgets. Confirm promising contrasts with three new paired seeds. Attribute a component only when its removal worsens validation loss in all confirmation pairs; interactions remain possible.

Transfer. Evaluate the selected recipe on a separate broader corpus, longer contexts and an additional structured task. Select that task and its success threshold before training. Require lower validation loss and at least 10% less median time to a predeclared quality target across three fresh pairs before advancing the scale claim. Measure inference latency and total stored state alongside training cost. A smaller numerical and resource pilot precedes every workload increase.

Training efficiency and inference latency remain separate measurements across all stages. Refinement proceeds as a separate research extension, with exact-solution gains across passes required before it becomes a primary capability claim.

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